

Programme	B. Sc. Statistics				
Course Code	STA4CJ202(P)				
Course Title	Bivariate Random Variables and Limit Theorems				
Type of Course	Major				
Semester	IV				
Academic Level	200-299				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	HSE level Mathematics/Statistics courses				
Course Summary Objective	Make students to aware bivariate distributions and understanding BVN. Apply LLN for computing asymptotic probability				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Calculate Marginal & Conditional Probability	U	C	Instructor-created exams / Quiz
CO2	Examine Independence of two Random Variables	Ap	P	Practical Assignment / Observation of Practical Skills
CO3	Examine properties of Bivariate Normal Distribution	Ap	P	Seminar Presentation / Group Tutorial Work
CO4	Compute upper and lower bound of Probability	U	C	Instructor-created exams / Home Assignments
CO5	Discriminate sequences of Random Variables satisfying Law of Large Numbers	Ap	P	One Minute Reflection Writing assignments
CO6	Asymptotic behavior of Random Variables	Ap	P	Viva Voce
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I	Bivariate Random Variable		10	18
	1	Joint probability mass function		
	2	Joint Probability density function		
	3	Marginal Probability functions		
	4	Conditional Probability functions		
	5	Joint Probability Distribution function		
	6	Properties of Joint Probability Distribution function		
	7	Independence of Random Variables.		
	Sections from References:			
II	Bivariate Expectation		11	18
	8	Mathematical expectation of Bivariate Random Variables,		
	9	Addition theorem of Expectation		
	10	Multiplication theorem of Expectation,		
	11	Covariance, Cauchy-Schwartz Inequality		
	12	Conditional Expectation and Conditional Variance		
	Sections from References:			
III	Bivariate Normal Distribution (BVN)		10	14
	13	Probability density function of BVN, properties of BVN		
	14	Marginal Probability density function of BVN		
	15	Conditional Probability density function of BVN		
	16	Standard bivariate normal distribution		
	Sections from References:			
IV	Limit Theorems		14	20
	17	Convergence in probability		
	18	Convergence in distribution		
	19	Chebyshev's Inequality		
	20	Weak Law of Large Numbers (iid case)		
	21	Bernoulli's Law of Large Numbers.		
	22	Central Limit Theorem (Lindberg levy-iid case),		
	23	Applications of CLT		
	24	Computation of sample size using Chebeshev's Inequality and CLT		
	Sections from References:			
V	Open Ended Module: Practical Applications		30	
	1	Hands-on in R: joint probability law, marginal and conditional probability functions, conditional expectation and variance, Chebyshev's inequality, WLLN, BVN		
	Sections from References:			
Books and References:				
1. S. C. Gupta and V. K. Kapoor. Fundamentals of Mathematical Statistics. Sultan Chand and Sons.				
2. Samuel Kotz, N. Balakrishnan, Norman L. Johnson. Continuous Multivariate Distributions: Models and Applications. Wiley Series in Probability and Statistics				